

```

CLR SCL ; TRANSITION SCL HIGH-TO-LOW
ACALL DEELAY ; DELAY FOR AN OPERATION
RET
; SUB READS THE CLOCK AND WRITES IT TO THE SCRATCHPAD MEMORY
; ON RETURN FROM HERE DATE & TIME DATA WILL BE STORED IN THE
; DATE & TIME REGISTERS FROM 24H (SECS) TO 24H (YEAR)
; ALARM SETTINGS IN REGISTERS 2CH(HRS) AND 2DH(MINUTES).
;*****
READ_CLOCK:
MOV R1,#SECS ; SECONDS
                STORAGE LOCATION
MOV BYTECNT,#00H
CLR LASTREAD
CALL SEND_START
MOV A,#DS1307H
CALL SEND_BYTE
MOV A,#00H
CALL SEND_BYTE
CALL SEND_STOP
CALL SEND_START
MOV A,#DS1307H
CALL SEND_BYTE
;*****
READ_LOOP:
MOV A,BYTECNT
CJNE A,#09H,NOT_LAST
SETB LASTREAD
NOT_LAST:CALL READ_BYTE
MOV @R1,A
MOV A,BYTECNT
CJNE A,#00H,NOT_FIRST
MOV A,#R1
CLR ACC.7 ; ENSURE OSC
                BIT=0 (ENABLED)
MOV @R1,A
NOT_FIRST:
INC R1
BYTECNT
MOV A,BYTECNT
CJNE A,#0AH,READ_LOOP
CALL SEND_STOP
NOP
NOP
RET
;((((((((((((((((((((((((((((((((((((((((((((((((((((((((
; STORE THE TIME TO RTC CHIP
;))))))))))))))))))))))))))))))))))))))))))))))))))))
STORE_CLOCK:
LCALL SEND_START ; SEND 2WIRE
                START CONDITION
MOV A,#DS1307H ; LOAD DS1307
                WRITE COMMAND
LCALL SEND_BYTE ; SEND WRITE COMMAND
MOV A,#00H ; SET DS1307
                DATA POINTER TO BEGINNING
LCALL SEND_BYTE ; OF USER RAM
                00H
MOV A,ALAH ; WRITE ALARM
                TIME TO RAM SPACE
LCALL SEND_BYTE
MOV A,ALAMIN
LCALL SEND_BYTE
LCALL SEND_STOP ; SEND 2WIRE
                STOP CONTION
LCALL SEND_START ; SEND 2WIRE
                START CONDITION
MOV A,#DS1307H ; LOAD
                DS1307 WRITE COMMAND
LCALL SEND_BYTE ; SEND WRITE COMMAND
                MOV A,#01H ; SET DS1307
                DATA POINTER TO BEGINNING
LCALL SEND_BYTE ; OF 00H
MOV A,MIN ; Send min
LCALL SEND_BYTE
MOV A,HRS ; send hr
                setb ACC.6 ; 12 HR MODE
CLR ACC.5 ; AM/PM
                1=PM,0=AM
LCALL SEND_BYTE
MOV A,DAY ; Send Day
LCALL SEND_BYTE
MOV A,DATE1 ; Send date
LCALL SEND_BYTE
MOV A,MONTH ; Send month
LCALL SEND_BYTE
MOV A,YEAR ; Send yr
                mov a,#90h
                icall send_byte
                mov a,alahr
                icall send_byte
                mov a,alamin
                icall send_byte
                LCALL SEND_STOP ; SEND 2WIRE
                STOP CONTION
RET
;*****
DEELAY: MOV R1,#00CH
REP2: MOV R2,#0FFH
REP1: NOP
        DJNZ R2,REP1
        DJNZ R1,REP2
        RET

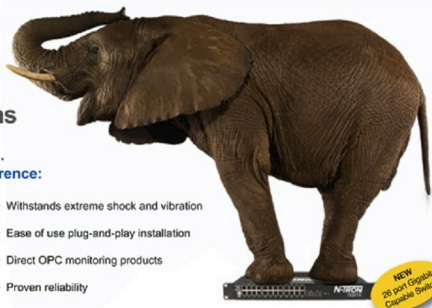
```

## Rugged enough to withstand the toughest conditions

Rugged, reliable, affordable... experience the **N-TRON** difference:

- Extended temperature range
- Highest noise immunity
- Withstands extreme shock and vibration
- Ease of use plug-and-play installation
- Direct OPC monitoring products
- Proven reliability

**N-TRON**  
THE INDUSTRIAL NETWORK COMPANY



NEW  
26 port Gigabit  
Capable Switch

Unmanaged  
Ethernet



Process  
Control



Media  
Converter



Managed  
Ethernet



Gigabit  
Ethernet



Wireless  
Access



Industrial  
PoE



For a free catalog visit [www.n-tron.com/catalog](http://www.n-tron.com/catalog) • +91 9844876540 • [N-TRON-India@n-tron.com](mailto:N-TRON-India@n-tron.com) • [www.n-tron.com](http://www.n-tron.com)